

Pesticide Exposure Assessment

- Using silicone wristbands in Chinese potato farmers -

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KEY MESSAGES

- Differences in pesticide exposure → agricultural practices **reduce pesticide exposure** among potato farmers
- Potato farmers in China use a multitude of pesticides: **14 different pesticide residues (PR)** on individual occupational wristbands

 42 farmers  2 weeks follow-up  Occupational exposure  Residential exposure

PROBLEM

China is the largest potato producer and consumer in the world.

Intensive potato production is associated with intensive pesticide use, a potential public health threat.

SOLUTION

Assessing pesticide exposure of Chinese potato farmers using silicone wristbands.



RELEVANCE

These findings could aid in educating farmers on how to minimize pesticide exposure during application.

Tabel 1. Number of detected PRs and cumulative PR ($\log \sum \text{PR}$) exposure on occupational (OCC) and residential (RES) wristbands of 42 Chinese potato farmers, in total and per agricultural group.

Agricultural group (N=42)	N PR OCC	N PR RES	Cumulative PR OCC (ng/WB) ($\log \sum \text{PR}$)	Cumulative PR RES (ng/WB) ($\log \sum \text{PR}$)
TOTAL Median (IQR)	14 (8)	3 (5)	3.9 (2)	2.3 (2)
Conventional Median (IQR)	15 (7)	4 (5)	3.9 (1)	2.5 (1)
Integrated Median (IQR)	16 (6)	5 (5)	4.5 (1)	2.6 (1)
Organic Median (IQR)	10 (8)	1 (3)	2.7 (1)	1.2 (2)